

Feedlot Economics and Management: A Visualization Tool for Decision Support

ABSTRACT (1 paragraph, ~350 words)

Briefly state the problem, objectives, methodology, main results, significance of the tool, and implications for stakeholders. *This will be written last.*

1. INTRODUCTION (4–5 paragraphs)

Context – global beef production, economic pressure on feedlots, complex decision-making environment.

Knowledge gap – traditional economic models and spreadsheet tools lack interactivity and intuitive use.

Emerging solutions – visual decision tools, system dynamics, or agent-based modeling for ag systems.

Objective of the paper – develop and test a visualization tool to support feedlot management decisions.

Paper structure overview.

2. BACKGROUND AND LITERATURE REVIEW (5–6 paragraphs)

Overview of feedlot economics – input costs, market variability, BRD, labor, margins.

Previous decision support tools – Excel-based, DSSAT, models, lacking real-time utility.

Gaps – limited integration of visual analytics, user-friendly dashboards, or risk-sensitive modeling.

Justification for a new approach that integrates economics and visualization.

- *Figure 1: Conceptual model of feedlot operations and key economic drivers.*

3. MATERIALS AND METHODS (5–6 paragraphs)

Overview of tool development framework (software used—Excel, Tableau, Power BI, etc.).

Economic variables and decision parameters modeled (feed costs, gains, feeder cost, etc.).

Visualization design – user interface, dashboard elements, interactivity, back-end calculations.

Validation approach – expert feedback, scenario testing, or comparison to historical data.

Description of user testing or pilot study (if available).

- *Figure 2:* Architecture or system design diagram of the visualization tool.
- *Figure 3:* Screenshot or mockup of the visualization dashboard.

4. RESULTS AND TOOL DEMONSTRATION (4–6 paragraphs)

Use-case scenarios – base case, high feed cost, cattle futures prediction, disease outbreak, etc.

Demonstrated outputs – profitability projections, break-even analysis, drought situations etc.

Comparative advantage – how the tool simplifies decision-making vs traditional models.

- *Figure 4:* Scenario simulation output chart (profitability over time under different assumptions).
- *Figure 5:* Sensitivity analysis visualization (tornado diagram, heatmap, etc.).

5. DISCUSSION (5–6 paragraphs)

Interpretation of results – what do the outputs mean for feedlot economics?

Decision support utility – how can different stakeholders (managers, vets, investors) benefit?

Practical implications – cost-saving, risk management, economic sustainability.

Tool limitations – data quality, scalability, regional specificity.

Integration potential – how it could work with IoT, ML, or herd management systems.

6. CONCLUSION AND FUTURE WORK (2–3 paragraphs)

Summary of contributions – novel tool, economic modeling, visualization benefits.

Next steps – tool expansion, integration with real-time data, user testing with commercial feedlots.

Policy relevance or scaling for sustainability metrics.

7. REFERENCES

citation format (APA style)

APPENDICES

- Tool screenshots
- Extended equations
- User instructions or link to demo version